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(54) ANTENNA DEVICE

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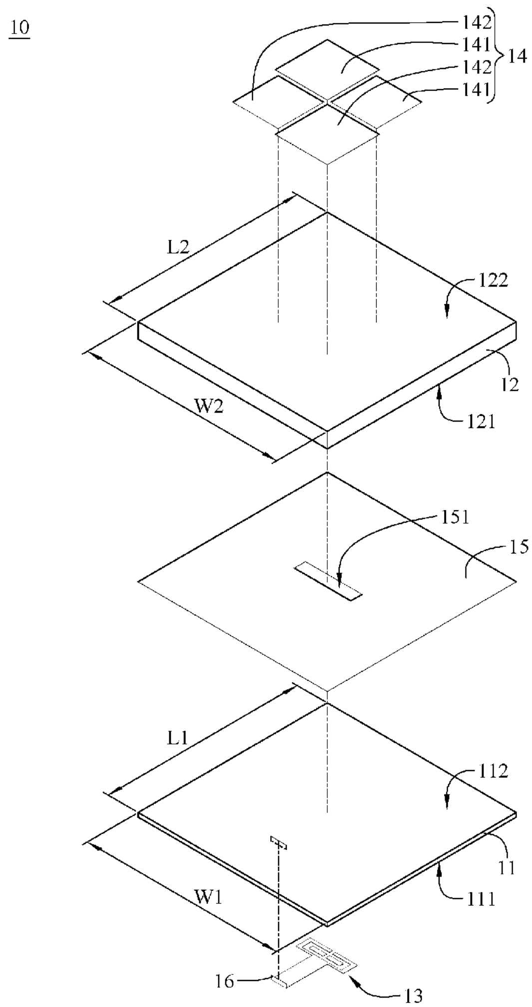
(57) **ABSTRACT**

An antenna device includes a first substrate, a second substrate, a T-shaped exciter and a radiator. The first substrate has a first bottom surface and a first top surface opposite to each other. The second substrate is stacked on the first substrate, and has a second bottom surface and a second top surface opposite to each other. The second bottom surface faces the first top surface. The T-shaped exciter is disposed on the first bottom surface, and includes a head portion and an extending portion. The extending portion is connected to a side of the head portion. The radiator is disposed on the second top surface, and includes two first radiation components and two second radiation components arranged along two diagonals of the radiator, respectively. A size of each of the two first radiation components is different from a size of each of the two second radiation components.



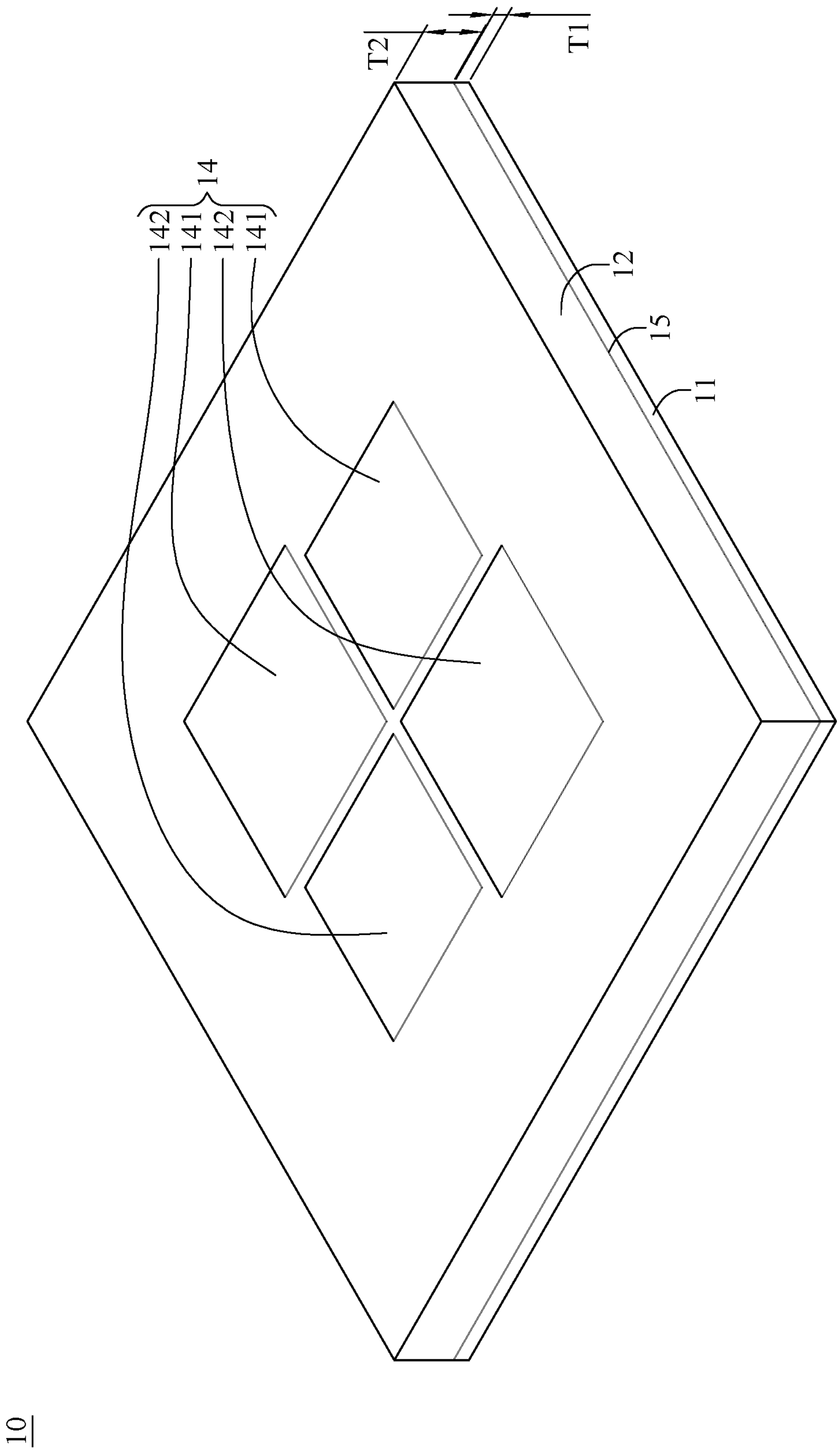


FIG. 1

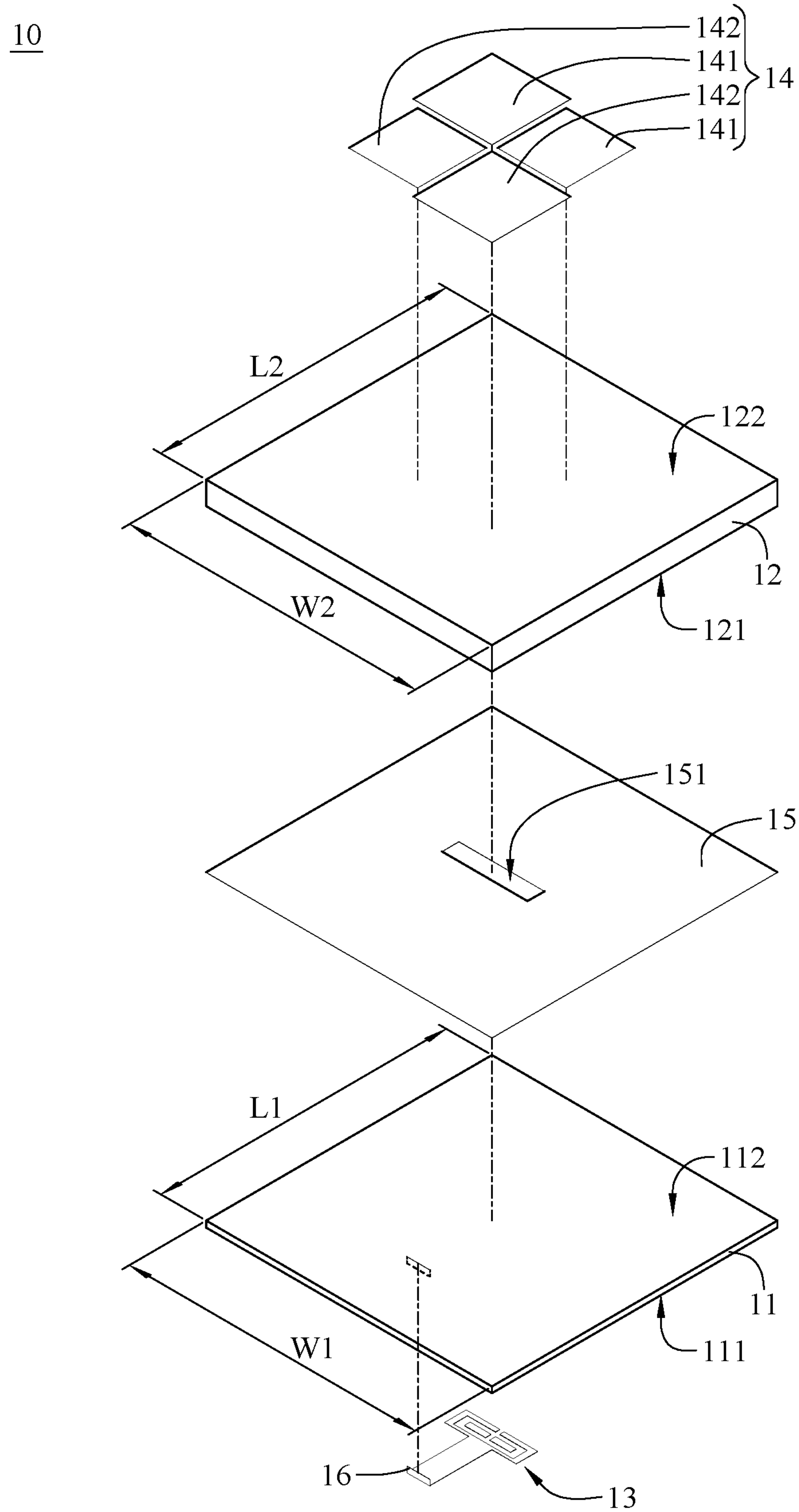
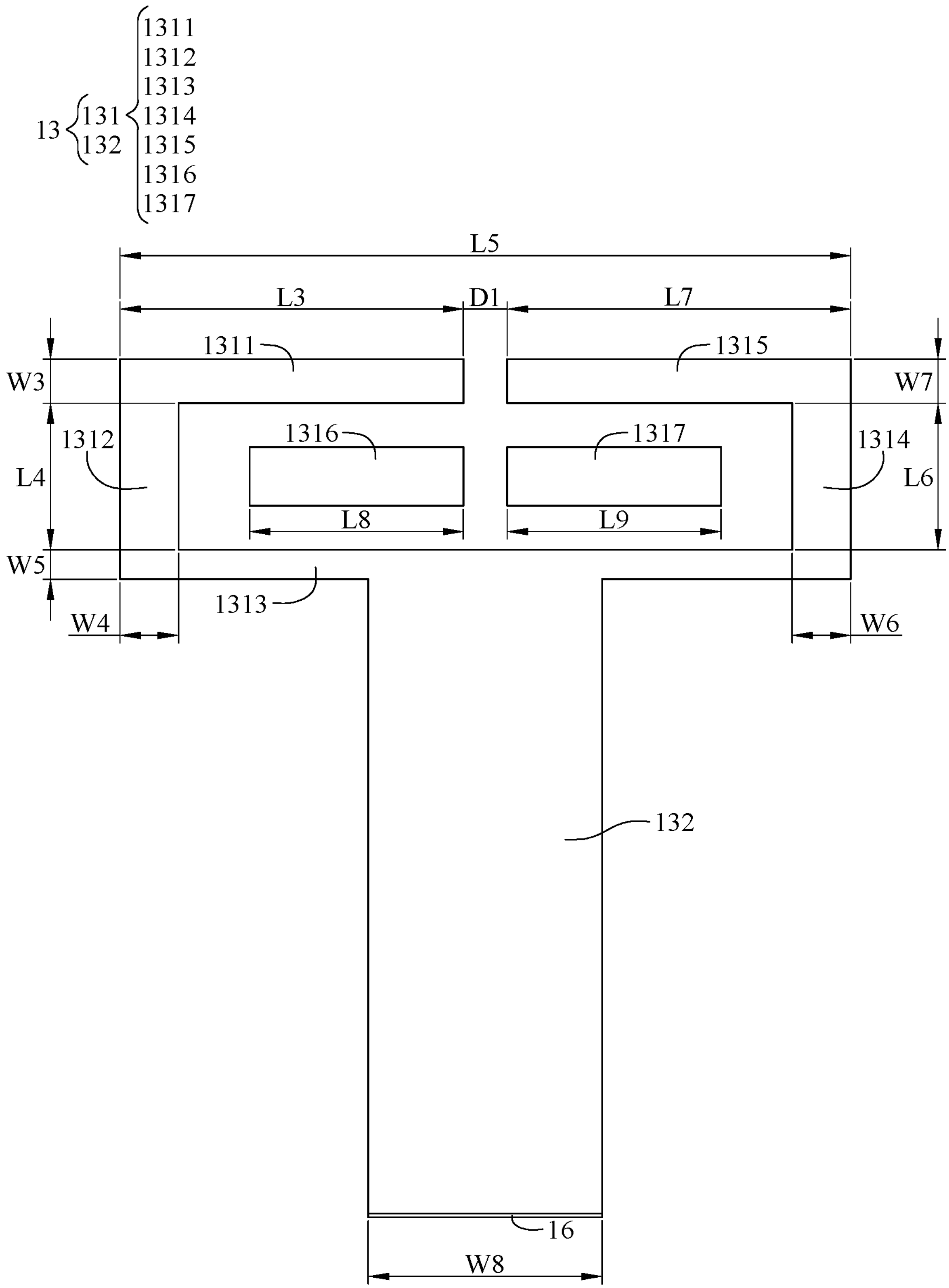


FIG. 2



13
131
132
1311
1312
1313
1314
1315
1316
1317
L5
L3
D1
L7
W3
W7
1311
1315
1312
1314
1316
1317
L4
L6
L8
L9
W5
1313
W4
W6
132
16
W8

FIG. 3

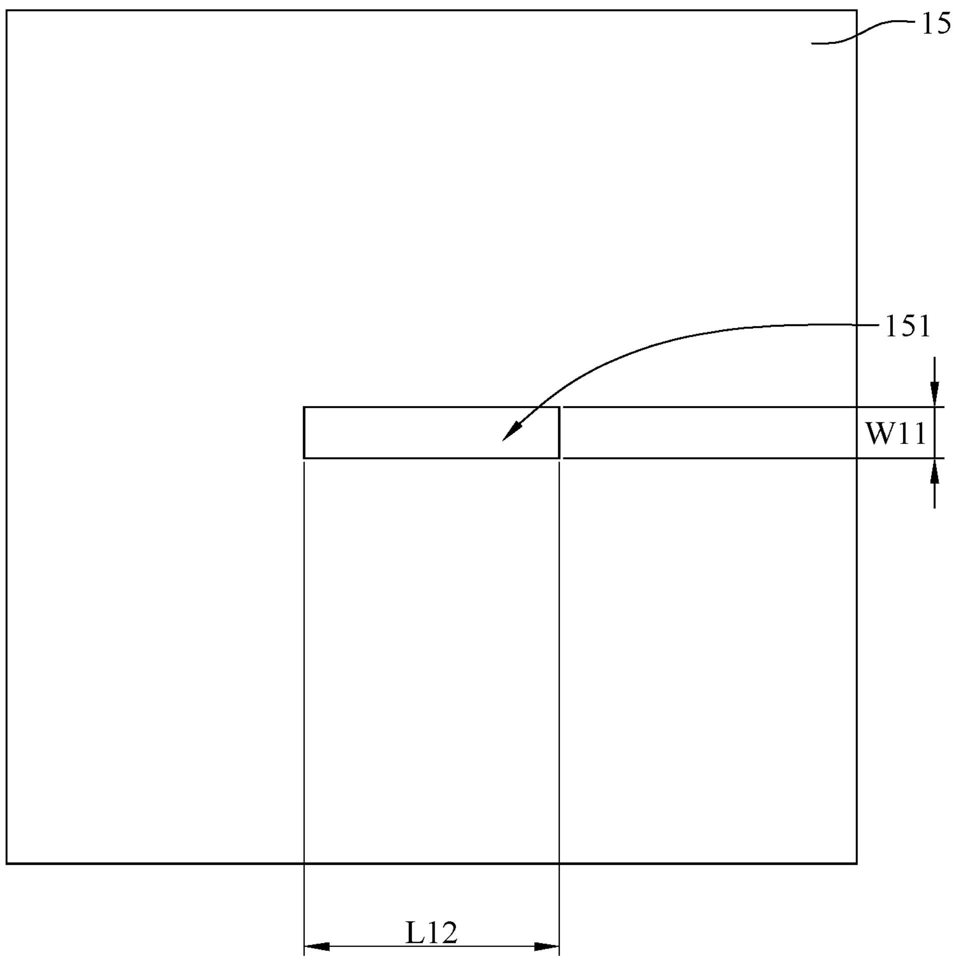


FIG. 4

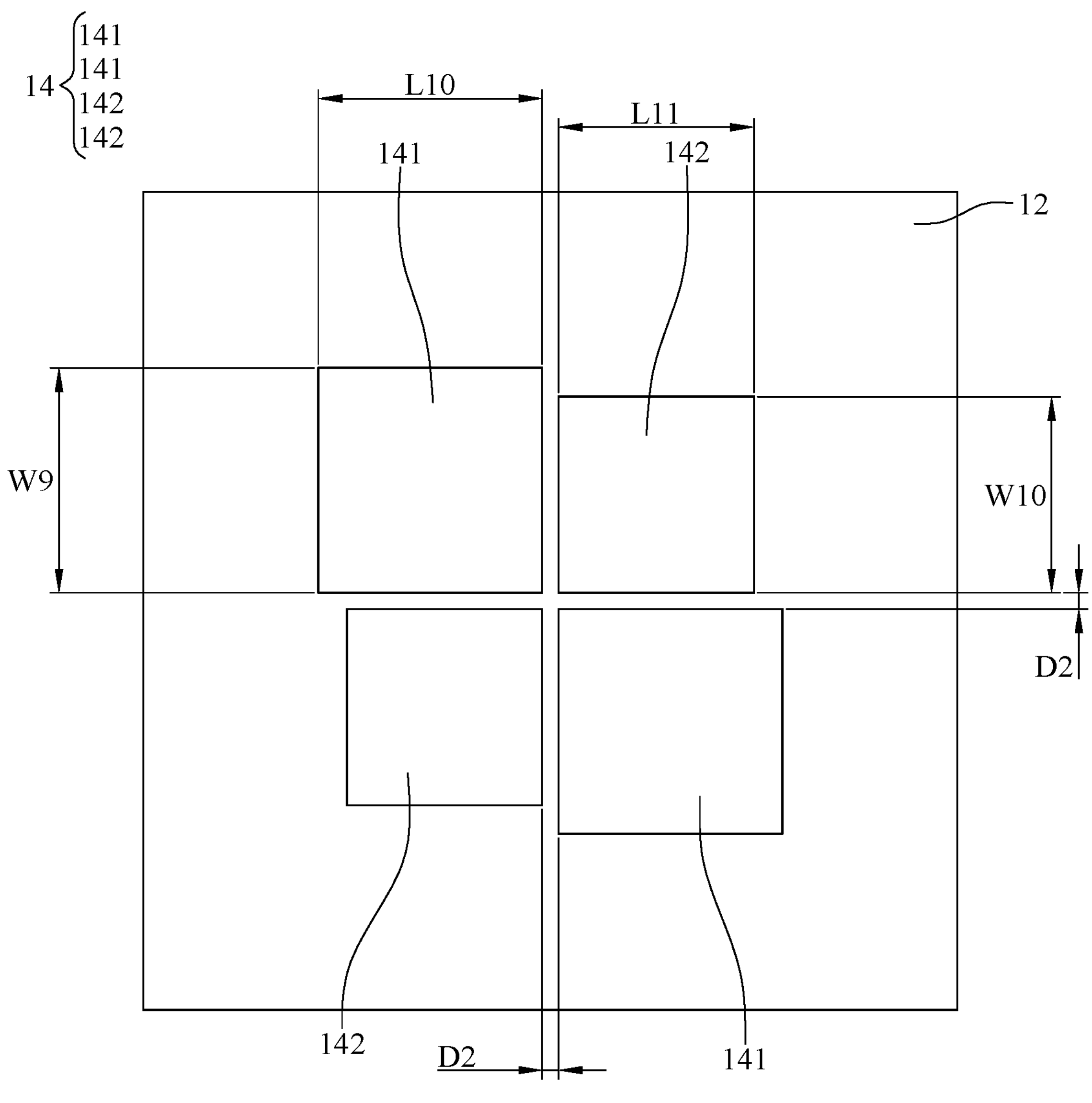


FIG. 5

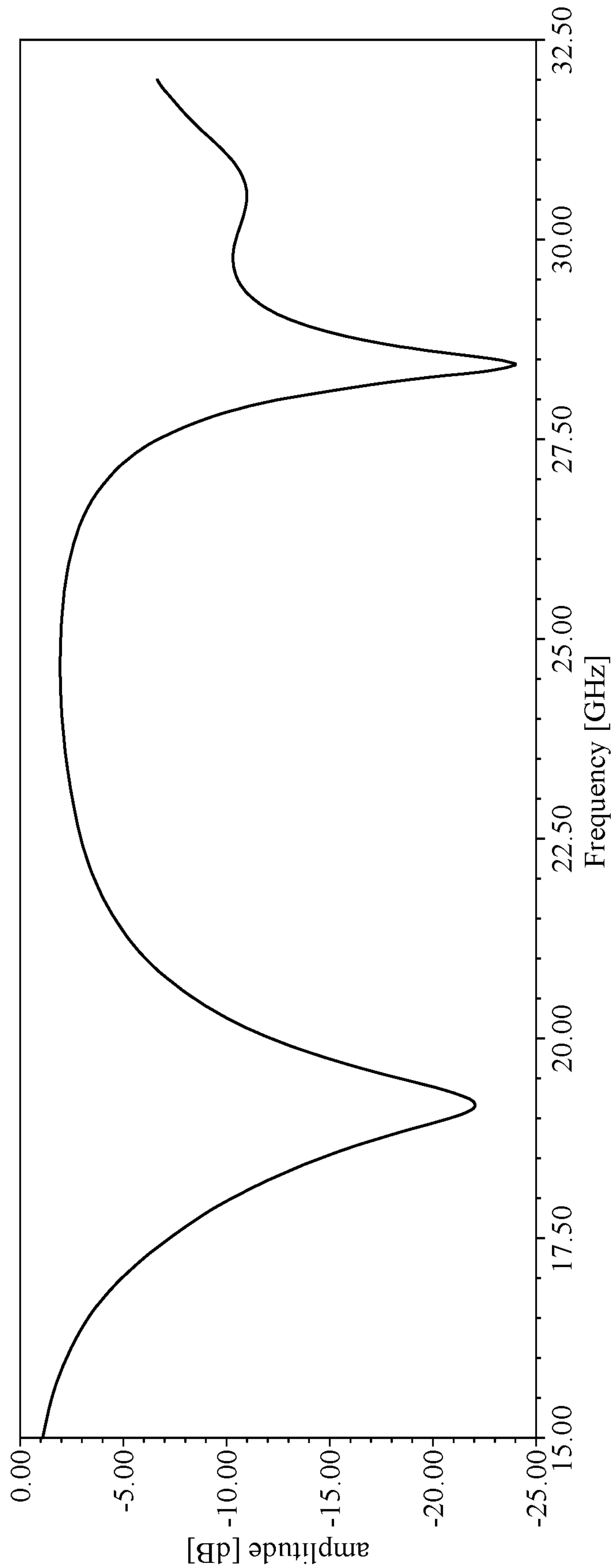

amplitude [dB]
0.00
-5.00
-10.00
-15.00
-20.00
-25.00
15.00
17.50
20.00
22.50
25.00
27.50
30.00
32.50
Frequency [GHz]

FIG. 6

10A

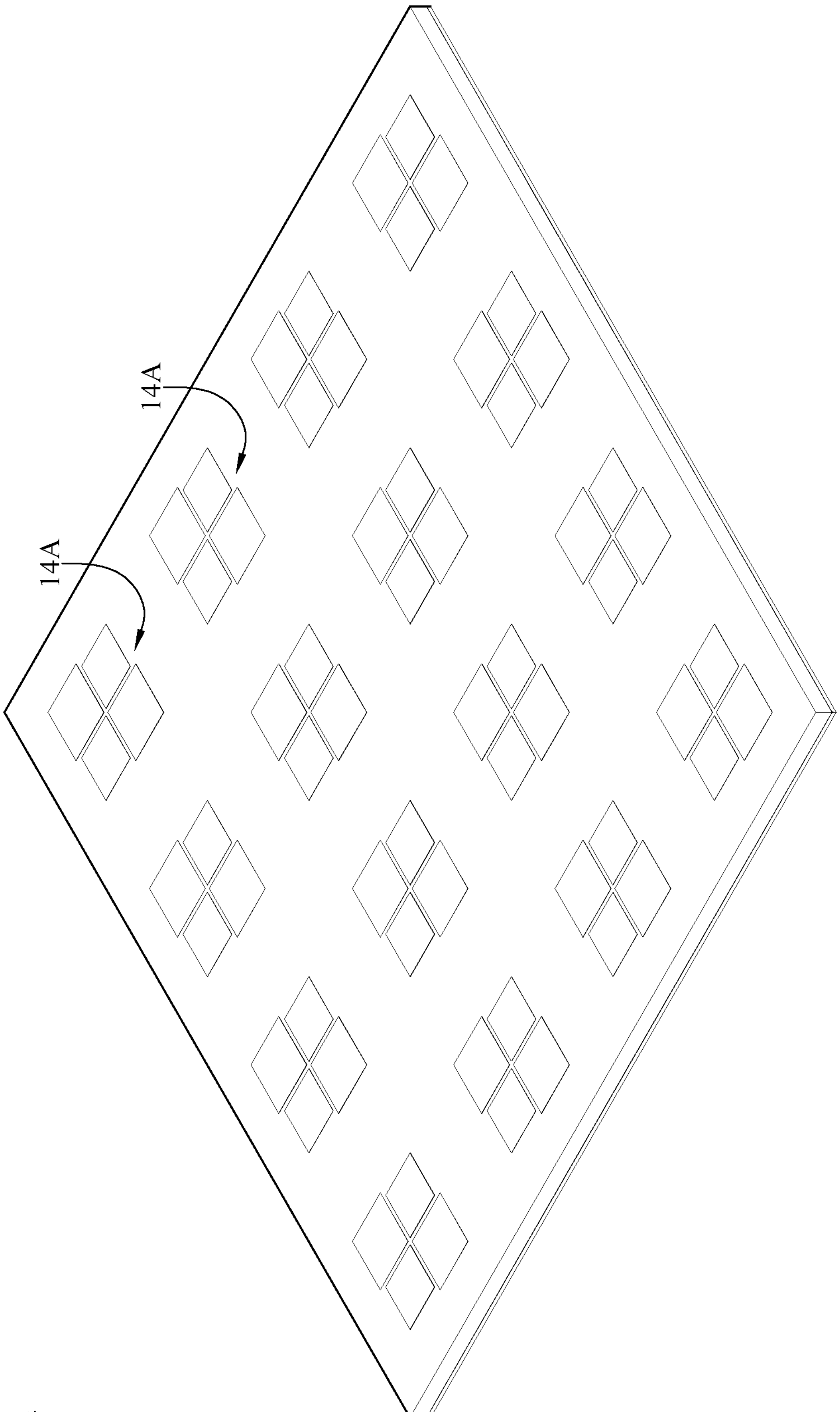


FIG. 7

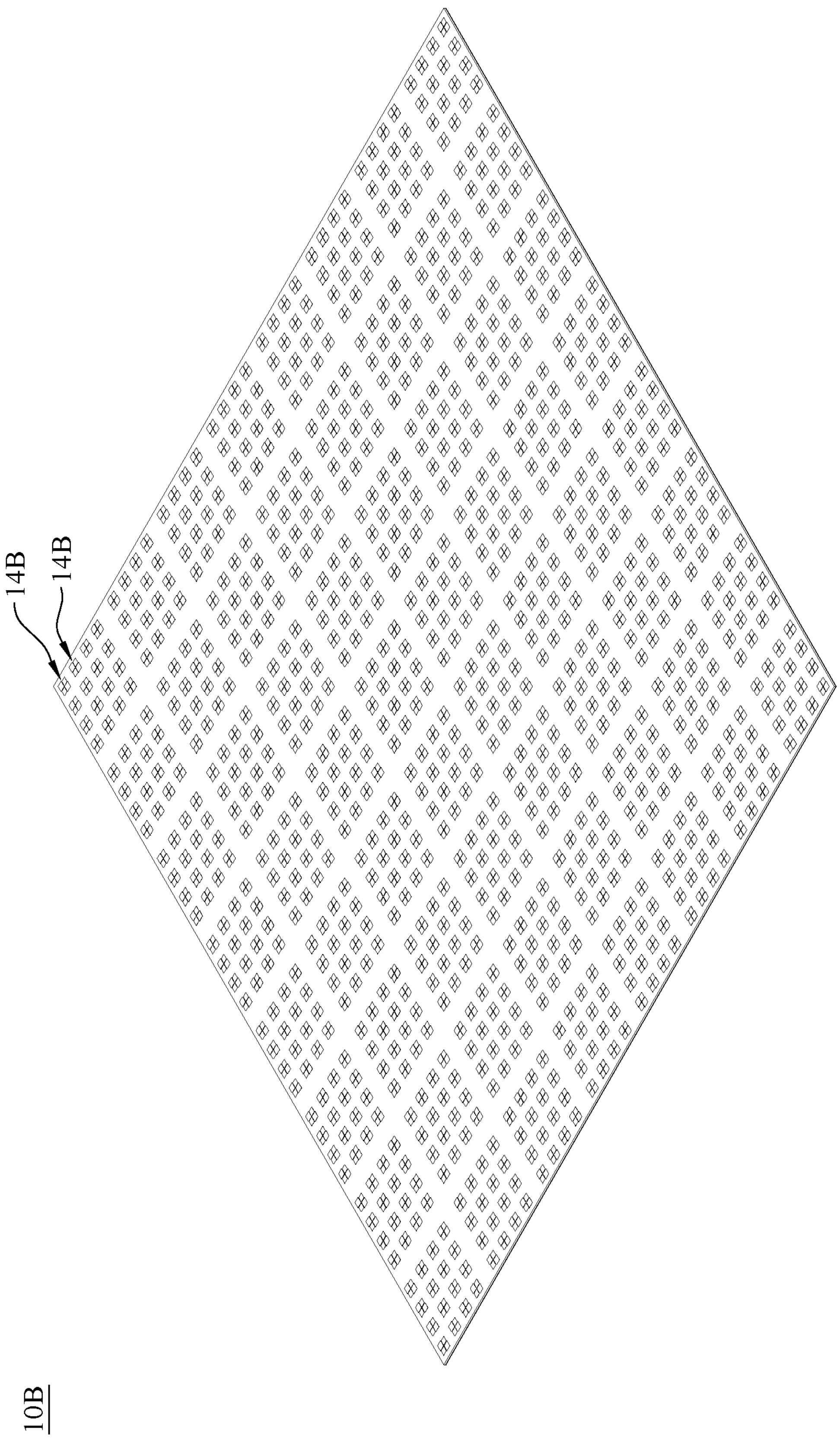


FIG. 8

ANTENNA DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No(s). 202410238917.9 filed in China, on Mar. 1, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Technical Field of the Invention

[0002] The invention relates to an antenna device, more particularly to an antenna device including radiator arranged along diagonals.

Description of the Related Art

[0003] With the advancement of wireless communication technology, electronic devices such as mobile phones and personal digital assistants are evolving towards more diversified functions, becoming lighter and thinner, and achieving faster and more efficient data transmission. In particular, wireless communication technology is expected to enter the 6G era by 2030, meeting various life applications and business requirements that 5G has not fulfilled. Specifically, low Earth orbit satellite communication is a key technology for the future commercialization of 6G.

[0004] Generally, ground base station antennas serve as the pivotal link between low Earth orbit (LEO) satellite systems and terrestrial communications, converting signals transmitted by satellites into usable data. The antennas are crucial for ensuring stable and efficient communication between ground facilities and LEO satellite. However, the current stability and efficiency of communications between 6G antennas and LEO satellites are still lacking. Therefore, improving the stability and efficiency of communications between antennas and LEO satellites is one of the key challenges that researchers need to address.

SUMMARY OF THE INVENTION

[0005] The invention provides an antenna device so as to improve the stability and efficiency of communications between antenna devices and LEO satellites.

[0006] One embodiment of the invention provides an antenna device including a first substrate, a second substrate, a T-shaped exciter and at least one radiator. The first substrate has a first bottom surface and a first top surface opposite to each other. The second substrate is stacked on the first substrate. The second substrate has a second bottom surface and a second top surface opposite to each other. The second bottom surface faces the first top surface. The T-shaped exciter is disposed on the first bottom surface. The T-shaped exciter includes a head portion and an extending portion. The extending portion is connected to a side of the head portion. The at least one radiator is disposed on the second top surface. The at least one radiator includes two first radiation components and two second radiation components. A size of each of the two first radiation components is different from a size of each of the two second radiation components. The two first radiation components are arranged along a diagonal of the at least one radiator. The two second radiation components are arranged along another diagonal of the at least one radiator.

[0007] According to the antenna device disclosed by the above embodiment, the size of each of the two first radiation components is different from the size of each of the two second radiation components, and the two first radiation components and the two second radiation components are arranged along the diagonals, respectively. Accordingly, in the two frequency bands ranging from 17.7 gigahertz (GHz) to 20.2 GHz and from 27.5 GHz to 30 GHz, the return losses of the antenna device are low. Therefore, the stability and efficiency of communications between the antenna device and low Earth orbit satellites can be improved.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The invention will become more fully understood from the detailed description given herein below and the accompanying drawings which are given by way of illustration only and thus are not limitative of the invention and wherein:

[0009] FIG. **1** is a perspective view of an antenna device in accordance with first embodiment of the invention;

[0010] FIG. **2** is an exploded view of the antenna device in FIG. **1**;

[0011] FIG. **3** is a plane view of a T-shaped exciter of the antenna device in FIG. **1**;

[0012] FIG. **4** is a plane view of a coupling body of the antenna device in FIG. **1**;

[0013] FIG. **5** is a plane view of a second substrate and a radiator of the antenna device in FIG. **1**;

[0014] FIG. **6** is a graph showing a return loss data of the antenna device in FIG. **1**;

[0015] FIG. **7** is a perspective view of an antenna device in accordance with second embodiment of the invention; and

[0016] FIG. **8** is a perspective view of an antenna device in accordance with third embodiment of the invention.

DETAILED DESCRIPTION

[0017] In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

[0018] In addition, the terms used in the invention, such as technical and scientific terms, have its own meanings and can be comprehended by those skilled in the art, unless the terms are additionally defined in the invention. That is, the terms used in the following paragraphs should be read on the meaning commonly used in the related fields and will not be overly explained, unless the terms have a specific meaning in the invention.

[0019] Please refer to FIG. **1** and FIG. **2**, where FIG. **1** is a perspective view of an antenna device **10** in accordance with first embodiment of the invention, and FIG. **2** is an exploded view of the antenna device **10** in FIG. **1**.

[0020] In this embodiment, the antenna device **10** is configured for, for example, low Earth orbit satellite communication, and includes a first substrate **11**, a second substrate **12**, a T-shaped exciter **13**, a radiator **14**, a coupling body **15** and a feeding point **16**. The first substrate **11** has a first bottom surface **111** and a first top surface **112** opposite to each other. The second substrate **12** is stacked on the first substrate **11**. The second substrate **12** has a second bottom surface **121** and a second top surface **122** opposite to each other. The second bottom surface **121** faces the first top surface **112**.

[0021] In this embodiment, the first substrate **11** and the second substrate **12** are, for example, made of RO4003 glass fiber materials, so that a loss of a signal radiated by the antenna device **10** can be reduced by the RO4003 glass fiber materials.

[0022] In this embodiment, a length L**1** of the first substrate **11**, a width W**1** of the first substrate **11**, a length L**2** of the second substrate **12** and a width W**2** of the second substrate **12** are each, for example, 10 millimeters. A thickness T**1** of the first substrate **11** is, for example, 0.8 millimeters. A thickness T**2** of the second substrate **12** is, for example, 0.2 millimeters.

[0023] Please refer to FIG. **1** and FIG. **5**, where FIG. **3** is a plane view of the T-shaped exciter **13** of the antenna device **10** in FIG. **1**, FIG. **4** is a plane view of the coupling body **15** of the antenna device **10** in FIG. **1**, and FIG. **5** is a plane view of the second substrate **12** and the radiator **14** of the antenna device **10** in FIG. **1**.

[0024] The T-shaped exciter **13** is disposed on the first bottom surface **111**, and is, for example, made of copper foil. The T-shaped exciter **13** includes a head portion **131** and an extending portion **132**. The extending portion **132** is connected to a side of the head portion **131**.

[0025] In detail, the head portion **131** of the T-shaped exciter **13** has a hollow structure, and includes a first section **1311**, a second section **1312**, a third section **1313**, a fourth section **1314**, a fifth section **1315**, a sixth section **1316** and a seventh section **1317**. An end of the first section **1311** is connected to an end of the second section **1312**. Another end of the second section **1312** is connected to an end of the third section **1313**. The first section **1311** and the third section **1313** are located on a same side of the second section **1312**. Another end of the third section **1313** is connected to an end of the fourth section **1314**. The second section **1312** and the fourth section **1314** are located on a same side of the third section **1313**. Another end of the fourth section **1314** is connected to an end of the fifth section **1315**. The third section **1313** and the fifth section **1315** are located on a same side of the fourth section **1314**. Another end of the fifth section **1315** and another end of the first section **1311** are spaced apart from each other. The first section **1311**, the second section **1312**, the third section **1313**, the fourth section **1314** and the fifth section **1315** together surround the sixth section **1316** and the seventh section **1317**. The sixth section **1316** is located between the first section **1311** and the third section **1313**. The seventh section **1317** is located between the third section **1313** and the fifth section **1315**. The sixth section **1316** and the seventh section **1317** are spaced apart from each other. The extending portion **132** is connected to a midsection of the third section **1313**.

[0026] In this embodiment, the T-shaped exciter **13** can generate two frequency bands ranging from 17.7 gigahertz (GHz) to 20.2 GHz and from 27.5 GHz to 30 GHz through the design of the head portion **131** with a hollow structure and the sixth section **1316** and the seventh section **1317** being spaced apart from each other.

[0027] In this embodiment, a length L**3** of the first section **1311** is, for example, 1.175 millimeters. A width W**3** of the first section **1311** is, for example, 0.15 millimeters. A length L**4** of the second section **1312** is, for example, 0.5 millimeters. A width W**4** of the second section **1312** is, for example, 0.2 millimeters. A length L**5** of the third section **1313** is, for example, 2.5 millimeters. A width W**5** of the third section **1313** is, for example, 0.1 millimeters. A length L**6** of the fourth section **1314** is, for example, 0.5 millimeters. A width W**6** of the fourth section **1314** is, for example, 0.2 millimeters. A length L**7** of the fifth section **1315** is, for example, 1.175 millimeters. A width W**7** of the fifth section **1315** is, for example, 0.15 millimeters. A length L**8** of the sixth section **1316** and a length L**9** of the seventh section **1317** are, for example, 0.73 millimeters. A distance D**1** between the first section **1311** and the fifth section **1315** is, for example, 0.15 millimeters. A width W**8** of the extending portion **132** is, for example, 0.8 millimeters.

[0028] The radiator **14** is disposed on the second top surface **122**, and is, for example, made of copper foil. The radiator **14** includes two first radiation components **141** and two second radiation components **142**. A size of each of the two first radiation components **141** is, for example, different from a size of each of the two second radiation components **142**. For example, the size of each of the two first radiation components **141** is greater than the size of each of the two second radiation components **142**. The two first radiation components **141** with larger size are configured to generate and radiate low frequency signals. The two second radiation components **142** with smaller size are configured to generate and radiate high frequency signals. The two first radiation components **141** are arranged along a diagonal of the radiator **14**. The two second radiation components **142** are arranged along another diagonal of the radiator **14**. In addition, there is a distance between each of the two first radiation components **141** and each of the two second radiation components **142**. Thus, impedance matching of the antenna device **10** can be improved and the electromagnetic wave band-pass effect can be achieved through the arrangement of the two first radiation components **141** and the two second radiation components **142**.

[0029] In this embodiment, a length L**10** of each of the two first radiation components **141** and a width W9 of each of the two first radiation components **141** are, for example, 2.75 millimeters. A length L**11** of each of the two second radiation components **142** and a width W**10** of each of the two second radiation components **142** are, for example, 2.4 millimeters. A distance D**2** between each of the two first radiation components **141** and each of the two second radiation components **142** is, for example, 0.2 millimeters.

[0030] The coupling body **15** is disposed between the first top surface **112** and the second bottom surface **121**, and is, for example, made of copper foil. The feeding point **16** is disposed at an end of the extending portion **132** farthest away from the head portion **131**, and is configured for a signal to be fed into the antenna device **10**. The coupling body **15** has a slot **151**. The slot **151** is configured for coupling the signal to the radiator **14**. A length L**12** of the slot **151** is, for example, 3 millimeters. A width W**11** of the slot **151** is, for example, 0.6 millimeters.

[0031] Please refer to FIG. **1** to FIG. **6**, which is a graph showing a return loss data of the antenna device **10** in FIG. **1**. In this embodiment, in the two frequency bands ranging from 17.7 GHz to 20.2 GHz and from 27.5 GHz to 30 GHz, the return losses of the antenna device **10** are low. In detail, in the two aforementioned frequency bands, the return losses of the antenna device **10** are mostly lower than −10 dB, so that the impedance matching of the antenna device **10** is good. In addition, generally, the higher the gain of the antenna is, the more concentrated the radiation from the antenna becomes, allowing the signal radiated from the antenna to be transmitted farther in a specific direction. In this embodiment, in the two aforementioned frequency bands, the antenna device **10**, for example, has a maximum gain of 6.7 dBi.

[0032] In this embodiment, the size of each of the two first radiation components **141** and the size of each of the two second radiation components **142** are different, and the two first radiation components **141** and the two second radiation components **142** are arranged along the diagonals, respectively. In addition, the head portion **131** of the T-shaped exciter **13** has a hollow structure, and the sixth section **1316** and the seventh section **1317** of the T-shaped exciter **13** are spaced apart from each other. Thus, the T-shaped exciter **13** can generate the two frequency bands ranging from 17.7 GHz to 20.2 GHz and from 27.5 GHz to 30 GHz, and the return losses in the two frequency bands are low. Therefore, the stability and efficiency of communications between the antenna device **10** and low Earth orbit satellites can be improved.

[0033] Please refer to FIG. **7**, which is a perspective view of an antenna device **10**A in accordance with second embodiment of the invention. The antenna device **10**A of this embodiment is similar to the antenna device **10** of the first embodiment, the main difference between them will be described below, and the same parts between them can be referred to the aforementioned paragraphs with the reference to FIG. **1** to FIG. **6** and will not be repeatedly introduced hereinafter. In this embodiment, the antenna device **10**A includes **16** radiators **14**A, and the radiators **14**A are arranged in, for example, a 4×4 array. Accordingly, by arranging the radiators **14**A in an array, the antenna device **10**A can achieve a significant gain superposition effect. In other embodiment, the antenna device may include other numbers of radiators arranged in an array according to a requirement for different electronic products.

[0034] Please refer to FIG. **8**, which is a perspective view of an antenna device **10**B in accordance with third embodiment of the invention. The antenna device **10**B of this embodiment is similar to the antenna device **10** of the first embodiment, the main difference between them will be described below, and the same parts between them can be referred to the aforementioned paragraphs with the reference to FIG. **1** to FIG. **6** and will not be repeatedly introduced hereinafter. In this embodiment, the antenna device **10**B includes **1024** radiators **14**B, and the radiators **14**B are arranged in, for example, a 32×32 array. Accordingly, by arranging the radiators **14**B in an array, the antenna device **10**B can achieve a significant gain superposition effect. In other embodiment, the antenna device may include other numbers of radiators arranged in an array according to a requirement for different electronic products.

[0035] According to the antenna device disclosed by the above embodiments, the size of each of the two first radiation components is different from the size of each of the two second radiation components, and the two first radiation components and the two second radiation components are arranged along the diagonals, respectively. In addition, the head portion of the T-shaped exciter has a hollow structure, and the sixth section and the seventh section of the T-shaped exciter are spaced apart from each other. Thus, the T-shaped exciter can generate the two frequency bands ranging from 17.7 GHz to 20.2 GHz and from 27.5 GHz to 30 GHz, and the return losses in the two frequency bands are low. Therefore, the stability and efficiency of communications between the antenna device and low Earth orbit satellites can be improved.

[0036] It will be apparent to those skilled in the art that various modifications and variations can be made to the invention. It is intended that the specification and examples be considered as exemplary embodiments only, with the scope of the invention being indicated by the following claims.

What is claimed is:

1. An antenna device, comprising:

a first substrate, having a first bottom surface and a first top surface opposite to each other;

a second substrate, stacked on the first substrate, wherein the second substrate has a second bottom surface and a second top surface opposite to each other, and the second bottom surface faces the first top surface;

a T-shaped exciter, disposed on the first bottom surface, wherein the T-shaped exciter comprises a head portion and an extending portion, and the extending portion is connected to a side of the head portion; and

at least one radiator, disposed on the second top surface, wherein the at least one radiator comprises two first radiation components and two second radiation components, a size of each of the two first radiation components is different from a size of each of the two second radiation components, the two first radiation components are arranged along a diagonal of the at least one radiator, and the two second radiation components are arranged along another diagonal of the at least one radiator.

2. The antenna device according to claim **1**, wherein the head portion of the T-shaped exciter comprises a first section, a second section, a third section, a fourth section, a fifth section, a sixth section and a seventh section, an end of the first section is connected to an end of the second section, another end of the second section is connected to an end of the third section, the first section and the third section are located on a same side of the second section, another end of the third section is connected to an end of the fourth section, the second section and the fourth section are located on a same side of the third section, another end of the fourth section is connected to an end of the fifth section, the third section and the fifth section are located on a same side of the fourth section, another end of the fifth section and another end of the first section are spaced apart from each other, the first section, the second section, the third section, the fourth section and the fifth section together surround the sixth section and the seventh section, the sixth section is located between the first section and the third section, the seventh section is located between the third section and the fifth section, the sixth section and the seventh section are spaced apart from each other, and the extending portion is connected to a midsection of the third section.

3. The antenna device according to claim **2**, wherein a length of the first section is 1.175 millimeters, a width of the first section is 0.15 millimeters, a length of the second section is 0.5 millimeters, a width of the second section is 0.2 millimeters, a length of the third section is 2.5 millimeters, a width of the third section is 0.1 millimeters, a length of the fourth section is 0.5 millimeters, a width of the fourth section is 0.2 millimeters, a length of the fifth section is

1.175 millimeters, a width of the fifth section is 0.15 millimeters, a length of the sixth section and a length of the seventh section are 0.73 millimeters, a distance between the first section and the fifth section is 0.15 millimeters, and a width of the extending portion is 0.8 millimeters.

4. The antenna device according to claim **1**, further comprising a coupling body and a feeding point, wherein the coupling body is disposed between the first top surface and the second bottom surface, the feeding point is disposed at an end of the extending portion farthest away from the head portion, the feeding point is configured for a signal to be fed into the antenna device, the coupling body has a slot, and the slot is configured for coupling the signal to the at least one radiator.

5. The antenna device according to claim **4**, wherein a length of the slot is 3 millimeters, and a width of the slot is 0.6 millimeters.

6. The antenna device according to claim **4**, wherein the T-shaped exciter, the at least one radiator and the coupling body are made of copper foil.

7. The antenna device according to claim **1**, wherein the antenna device comprises a plurality of radiators, and the plurality of radiators are arranged in an array.

8. The antenna device according to claim **1**, wherein a length of the first substrate, a width of the first substrate, a length of the second substrate and a width of the second substrate are 10 millimeters, a thickness of the first substrate is 0.8 millimeters, and a thickness of the second substrate is 0.2 millimeters.

9. The antenna device according to claim **1**, wherein a length of each of the two first radiation components and a width of each of the two first radiation components are 2.75 millimeters, a length of each of the two second radiation components and a width of each of the two second radiation components are 2.4 millimeters, and a distance between each of the two first radiation components and each of the two second radiation components is 0.2 millimeters.

10. The antenna device according to claim **1**, wherein the first substrate and the second substrate are made of glass fiber materials.

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